

Tangent half-angle formula

In trigonometry, **tangent half-angle formulas** relate the tangent of half of an angle to trigonometric functions of the entire angle.^[1]

Formulae

The tangent of half an angle is the stereographic projection of the circle through the point at angle π radians onto the line through the angles $\pm \frac{\pi}{2}$. Tangent half-angle formulae include

$$\tan \frac{1}{2}(\eta \pm \theta) = \frac{\tan \frac{1}{2}\eta \pm \tan \frac{1}{2}\theta}{1 \mp \tan \frac{1}{2}\eta \tan \frac{1}{2}\theta} = \frac{\sin \eta \pm \sin \theta}{\cos \eta + \cos \theta} = -\frac{\cos \eta - \cos \theta}{\sin \eta \mp \sin \theta},$$

with simpler formulae when η is known to be 0, $\pi/2$, π , or $3\pi/2$ because $\sin(\eta)$ and $\cos(\eta)$ can be replaced by simple constants.

In the reverse direction, the formulae include

$$\sin \alpha = \frac{2 \tan \frac{1}{2}\alpha}{1 + \tan^2 \frac{1}{2}\alpha}$$

$$\cos \alpha = \frac{1 - \tan^2 \frac{1}{2}\alpha}{1 + \tan^2 \frac{1}{2}\alpha}$$

$$\tan \alpha = \frac{2 \tan \frac{1}{2}\alpha}{1 - \tan^2 \frac{1}{2}\alpha}.$$

Proofs

Algebraic proofs

Using the angle addition and subtraction formulae for both the sine and cosine one obtains

$$\sin(a + b) + \sin(a - b) = 2 \sin a \cos b$$

$$\cos(a + b) + \cos(a - b) = 2 \cos a \cos b.$$

Setting $a = \frac{1}{2}(\eta + \theta)$ and $b = \frac{1}{2}(\eta - \theta)$ and substituting yields

$$\sin \eta + \sin \theta = 2 \sin \frac{1}{2}(\eta + \theta) \cos \frac{1}{2}(\eta - \theta)$$

$$\cos \eta + \cos \theta = 2 \cos \frac{1}{2}(\eta + \theta) \cos \frac{1}{2}(\eta - \theta).$$

Dividing the sum of sines by the sum of cosines gives

$$\frac{\sin \eta + \sin \theta}{\cos \eta + \cos \theta} = \tan \frac{1}{2}(\eta + \theta).$$

Also, a similar calculation starting with $\sin(a + b) - \sin(a - b)$ and $\cos(a + b) - \cos(a - b)$ gives

$$-\frac{\cos \eta - \cos \theta}{\sin \eta - \sin \theta} = \tan \frac{1}{2}(\eta + \theta).$$

Furthermore, using double-angle formulae and the Pythagorean identity $1 + \tan^2 \frac{1}{2}\alpha = 1/\cos^2 \frac{1}{2}\alpha$ gives

$$\begin{aligned}\sin \alpha &= 2 \sin \frac{1}{2}\alpha \cos \frac{1}{2}\alpha = \frac{2 \sin \frac{1}{2}\alpha \cos \frac{1}{2}\alpha / \cos^2 \frac{1}{2}\alpha}{1 + \tan^2 \frac{1}{2}\alpha} = \frac{2 \tan \frac{1}{2}\alpha}{1 + \tan^2 \frac{1}{2}\alpha} \\ \cos \alpha &= \cos^2 \frac{1}{2}\alpha - \sin^2 \frac{1}{2}\alpha = \frac{(\cos^2 \frac{1}{2}\alpha - \sin^2 \frac{1}{2}\alpha) / \cos^2 \frac{1}{2}\alpha}{1 + \tan^2 \frac{1}{2}\alpha} = \frac{1 - \tan^2 \frac{1}{2}\alpha}{1 + \tan^2 \frac{1}{2}\alpha}.\end{aligned}$$

Taking the quotient of the formulae for sine and cosine yields

$$\tan \alpha = \frac{2 \tan \frac{1}{2}\alpha}{1 - \tan^2 \frac{1}{2}\alpha}.$$

Geometric proofs

Applying the formulae derived above to the rhombus figure on the right, it is readily shown that

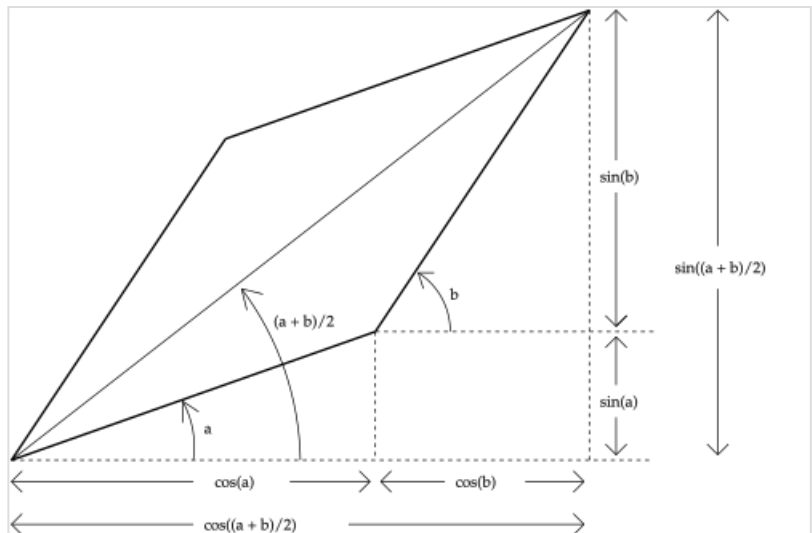
$$\tan \frac{1}{2}(a + b) = \frac{\sin \frac{1}{2}(a + b)}{\cos \frac{1}{2}(a + b)} = \frac{\sin a + \sin b}{\cos a + \cos b}.$$

In the unit circle, application of the above shows that $t = \tan \frac{1}{2}\varphi$. By similarity of triangles,

$$\frac{t}{\sin \varphi} = \frac{1}{1 + \cos \varphi}.$$

It follows that

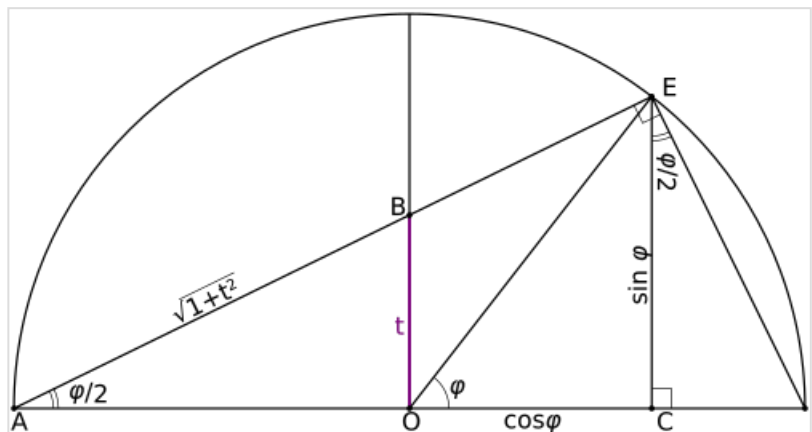
$$t = \frac{\sin \varphi}{1 + \cos \varphi} = \frac{\sin \varphi(1 - \cos \varphi)}{(1 + \cos \varphi)(1 - \cos \varphi)} = \frac{1 - \cos \varphi}{\sin \varphi}.$$



The sides of this rhombus have length 1. The angle between the horizontal line and the shown diagonal is $\frac{1}{2}(a+b)$. This is a geometric way to prove the particular tangent half-angle formula that says $\tan \frac{1}{2}(a+b) = (\sin a + \sin b) / (\cos a + \cos b)$. The formulae $\sin \frac{1}{2}(a+b)$ and $\cos \frac{1}{2}(a+b)$ are the ratios of the actual distances to the length of the diagonal.

The tangent half-angle substitution in integral calculus

In various applications of trigonometry, it is useful to rewrite the trigonometric functions (such as sine and cosine) in terms of rational functions of a new variable t . These identities are known collectively as the **tangent half-angle formulae** because of the definition of t . These identities can be useful in calculus for converting rational functions in sine and cosine to functions of t in order to find their antiderivatives.



A geometric proof of the tangent half-angle substitution

Geometrically, the construction goes like this: for any point $(\cos \phi, \sin \phi)$ on the unit circle, draw the line passing through it and the point $(-1, 0)$. This point crosses the y -axis at some point $y = t$. One can show using simple geometry that $t = \tan(\phi/2)$. The equation for the drawn line is $y = (1+x)t$. The equation for the intersection of the line and circle is then a quadratic equation involving t . The two solutions to this equation are $(-1, 0)$ and $(\cos \phi, \sin \phi)$. This allows us to write the latter as rational functions of t (solutions are given below).

The parameter t represents the stereographic projection of the point $(\cos \phi, \sin \phi)$ onto the y -axis with the center of projection at $(-1, 0)$. Thus, the tangent half-angle formulae give conversions between the stereographic coordinate t on the unit circle and the standard angular coordinate ϕ .

Then we have

$$\sin \varphi = \frac{2t}{1+t^2}, \quad \cos \varphi = \frac{1-t^2}{1+t^2},$$

$$\tan \varphi = \frac{2t}{1-t^2} \quad \cot \varphi = \frac{1-t^2}{2t},$$

$$\sec \varphi = \frac{1+t^2}{1-t^2}, \quad \csc \varphi = \frac{1+t^2}{2t},$$

and

$$e^{i\varphi} = \frac{1+it}{1-it}, \quad e^{-i\varphi} = \frac{1-it}{1+it}.$$

Both this expression of $e^{i\varphi}$ and the expression $t = \tan(\varphi/2)$ can be solved for φ . Equating these gives the arctangent in terms of the natural logarithm

$$\arctan t = \frac{-i}{2} \ln \frac{1+it}{1-it}.$$

In calculus, the tangent half-angle substitution is used to find antiderivatives of rational functions of $\sin \varphi$ and $\cos \varphi$. Differentiating $t = \tan \frac{1}{2} \varphi$ gives

$$\frac{dt}{d\varphi} = \frac{1}{2} \sec^2 \frac{1}{2} \varphi = \frac{1}{2} (1 + \tan^2 \frac{1}{2} \varphi) = \frac{1}{2} (1 + t^2)$$

and thus

$$d\varphi = \frac{2 dt}{1+t^2}.$$

Hyperbolic identities

One can play an entirely analogous game with the hyperbolic functions. A point on (the right branch of) a hyperbola is given by $(\cosh \psi, \sinh \psi)$. Projecting this onto y-axis from the center $(-1, 0)$ gives the following:

$$t = \tanh \frac{1}{2} \psi = \frac{\sinh \psi}{\cosh \psi + 1} = \frac{\cosh \psi - 1}{\sinh \psi}$$

with the identities

$$\sinh \psi = \frac{2t}{1-t^2}, \quad \cosh \psi = \frac{1+t^2}{1-t^2},$$

$$\tanh \psi = \frac{2t}{1+t^2}, \quad \coth \psi = \frac{1+t^2}{2t},$$

$$\operatorname{sech} \psi = \frac{1-t^2}{1+t^2}, \quad \operatorname{csch} \psi = \frac{1-t^2}{2t},$$

and

$$e^\psi = \frac{1+t}{1-t}, \quad e^{-\psi} = \frac{1-t}{1+t}.$$

Finding ψ in terms of t leads to following relationship between the inverse hyperbolic tangent **artanh** and the natural logarithm:

$$2 \operatorname{artanh} t = \ln \frac{1+t}{1-t}.$$

The hyperbolic tangent half-angle substitution in calculus uses

$$d\psi = \frac{2 dt}{1-t^2}.$$

The Gudermannian function

Comparing the hyperbolic identities to the circular ones, one notices that they involve the same functions of t , just permuted. If we identify the parameter t in both cases we arrive at a relationship between the circular functions and the hyperbolic ones. That is, if

$$t = \tan \frac{1}{2}\varphi = \tanh \frac{1}{2}\psi$$

then

$$\varphi = 2 \operatorname{arctan} \left(\tanh \frac{1}{2}\psi \right) \equiv \operatorname{gd} \psi.$$

where $\operatorname{gd}(\psi)$ is the Gudermannian function. The Gudermannian function gives a direct relationship between the circular functions and the hyperbolic ones that does not involve complex numbers. The above descriptions of the tangent half-angle formulae (projection the unit circle and standard hyperbola onto the y -axis) give a geometric interpretation of this function.

Rational values and Pythagorean triples

Starting with a Pythagorean triangle with side lengths a , b , and c that are positive integers and satisfy $a^2 + b^2 = c^2$, it follows immediately that each interior angle of the triangle has rational values for sine and cosine, because these are just ratios of side lengths. Thus each of these angles has a rational value for its half-angle tangent, using $\tan \varphi/2 = \sin \varphi / (1 + \cos \varphi)$.

The reverse is also true. If there are two positive angles that sum to 90° , each with a rational half-angle tangent, and the third angle is a right angle then a triangle with these interior angles can be scaled to a Pythagorean triangle. If the third angle is not required to be a right angle, but is the angle that makes the three positive angles sum to 180° then the third angle will necessarily have a rational number for its half-angle tangent when the first two do (using angle addition and subtraction formulas for tangents) and the triangle can be scaled to a Heronian triangle.

Generally, if K is a subfield of the complex numbers then $\tan \varphi/2 \in K \cup \{\infty\}$ implies that $\{\sin \varphi, \cos \varphi, \tan \varphi, \sec \varphi, \csc \varphi, \cot \varphi\} \subseteq K \cup \{\infty\}$.

See also



- List of trigonometric identities
- Half-side formula

External links

- "Tangent formula" (https://www.encyclopediaofmath.org/index.php?title=Tangent_formula), *Encyclopedia of Mathematics*, EMS Press, 2001 [1994]
- Tangent half-angle formula (<https://planetmath.org/TangentOfHalvedAngle>) at PlanetMath.

References

1. *Mathematics* (<https://books.google.com/books?id=ztcDUEiHLCYC&pg=SA6-PA19>). United States, NAVEDTRA [i.e. Naval] Education and Training Program Management Support Activity, 1989. 6-19.

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